KHALIL AYEB

DATA SCIENCE & AI ENGINEER

Mahdia, Tunisia (+216) 58 153 752 ayebkalil@gmail.com **Portfolio (Interactive Demos)** github.com/ayebkalil

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Data Science & AI Engineer specialized in Generative AI, RAG architectures, Multimodal Deep Learning, and Computer Vision. Track record of developing production ML pipelines, low-latency APIs, and interactive 3D/AI applications.

TECHNICAL SKILLS

Languages:	Python, SQL, TypeScript, JavaScript, Java (Spring Boot), C++, HTML5/CSS3
AI / Deep Learning:	PyTorch, Transformers (AraBERT), UNet++, cGAN (Pix2Pix), OpenCV, Scikit-learn, XGBoost, MediaPipe
GenAI & Retrieval:	GraphRAG, Hybrid Search (BM25 + Dense RRF), Neo4j, Qdrant, ChromaDB, FAISS, LangChain, Gemini API
Backend & Full-Stack:	FastAPI, Next.js 16, React 19, Three.js (WebGL), Django, RESTful APIs, WebSockets, PostgreSQL, DuckDB
MLOps & Tools:	Docker, MLflow, ONNX Runtime, DVC, Apache Parquet, MinIO, Redis, Git, Linux, CI/CD, Power BI

WORK EXPERIENCE

Software Engineering & Computer Vision Intern — IronPulse Health Labs — Aug 2026 – Sep 2026

Next.js 16, TypeScript, MediaPipe, WebGL, WebSockets, PostgreSQL, Google Fit API

- **Biometrics & Edge Vision:** Ingested 10,000+ daily biometric points via Google Fit API with sub-second sync latency; implemented in-browser 3D pose estimation (MediaPipe/WebGL) with 98.4% repetition accuracy and a live 1v1 WebSocket battle engine (<45ms latency).

AI & RAG Engineering Intern — 9anouni AI (GitHub) — Jun 2026 – Jul 2026

Python, FastAPI, PostgreSQL (btree_gist), ChromaDB, React 19, TypeScript, Google Gemini, Docker

- **Bitemporal Legal RAG:** Indexed 1,114 statutory articles in PostgreSQL 16 with GiST interval exclusion for amendment time-travel; engineered hybrid retrieval (BM25 + ChromaDB via RRF) hitting 95.2% Recall@5 (under 50ms) with Gemini defensive grounding.

NLP Research Intern (End-of-Studies PFE) — Faculty of Sciences of Monastir (FSM) — Feb 2024 – May 2024

Python, PyTorch, Hugging Face Transformers, AraBERT, Scikit-learn, Pandas

- **Arabic NER Transformer:** Fine-tuned AraBERT on 50,000+ annotated tokens for entity extraction (PER, ORG, LOC), achieving a 91.4% macro F1-score (+8.2% boost over baseline models) with automated morphological preprocessing.

ACADEMIC & ENGINEERING PROJECTS

[View Live Demos ↗]

OmniLake-GraphRAG — Multimodal Lakehouse & Hybrid GraphRAG (GitHub) — 2026

Python, FastAPI, React, Docker, Neo4j, Qdrant, Redis, DuckDB, Apache Parquet, MinIO, Ollama

- **Lakehouse & Graph AI:** Built a zero-cloud-cost financial Lakehouse fusing dense vector search with Neo4j PageRank (100% multi-hop reasoning accuracy); built an autonomous text-to-Cypher agent and a Redis vector cache slashing repeat latency to 42ms (89.5×).

ArchiGen 3D — Generative AI & BIM Procedural 3D Architecture Engine — 2026

Python, PyTorch (CUDA), FastAPI, OpenCV, Shapely, Three.js, React, Docker, ONNX Runtime, DVC

- **Blueprint-to-3D Synthesis:** Deployed MultiTask UNet++ (ResNet-34) for floorplan parsing, Pix2Pix cGAN for facades, and local TripoSR for 3D meshes on 4GB VRAM; engineered a procedural vectorization engine rendered live in Three.js (PBR/HDRI).

Multimodal Deep Learning for Longitudinal Parkinson's Prognosis — 2025 – 2026

Python, PyTorch, XGBoost, Scikit-learn, Streamlit, Pandas, PPMI Clinical & SPECT/MRI Datasets

- **Clinical Multimodal Fusion:** Unified longitudinal exams and 3D SPECT/MRI scans (5,426 patients) via Late Gated MoE, outperforming temporal baselines (MAE 5.158 pts, p = 0.0224); deployed a Streamlit dashboard simulating patient progression trajectories.

MentalHealth AI — Production Django ML Backend & Risk Prediction API — 2026

Python, Django, Django REST Framework, XGBoost, Scikit-learn, MongoDB (Djongo), Docker, Gunicorn

- **Multi-Model ML & Low-Latency API:** Trained a 5-model pipeline on 10,000 records; deployed XGBoost achieving 100% Recall on high-risk cases (99.7% CV score) across a Django REST API (<50ms latency) with MongoDB, SHA-256 anonymization, and batch scoring.

Competitive Game Analytics & Smurf Detection System — Feb 2026 – Apr 2026

Python, Scikit-learn, XGBoost, FastAPI, MLflow, Docker, Pandas

- **Player Telemetry Anomaly Detection:** Engineered feature pipelines and trained ML classification models on player telemetry to identify smurf accounts; tracked experiments with MLflow and packaged inference pipelines into Dockerized FastAPI services.

SafeTravel — AI-Powered Global Risk & Travel Safety Intelligence — 2025 – 2026

Java 21, Spring Boot 3, Spring Security (JWT), PostgreSQL, React 19, Three.js (WebGL), Gemini API, Zustand

- **3D Geospatial Intelligence:** Built an interactive WebGL 3D globe (Three.js) mapping safety ratings for 190+ countries with Gemini news classification; engineered a decay scoring engine in Spring Boot with stateless JWT/RBAC security (sub-100ms latency).

EDUCATION

Engineering Degree: Data Science & AI — 2024 – 2027
Tek-Up University — Ariana, Tunisia

B.S. in Software Engineering & Computer Systems — 2021 – 2024
Faculty of Sciences of Monastir (FSM) — Tunisia

CERTIFICATIONS & LANGUAGES

PCAP: Certified Associate in Python Programming — *Python Institute*

PL-300: Microsoft Power BI Data Analyst — *Microsoft Certified*

Languages: English (Fluent), French (Fluent), Arabic (Native / Bilingual)